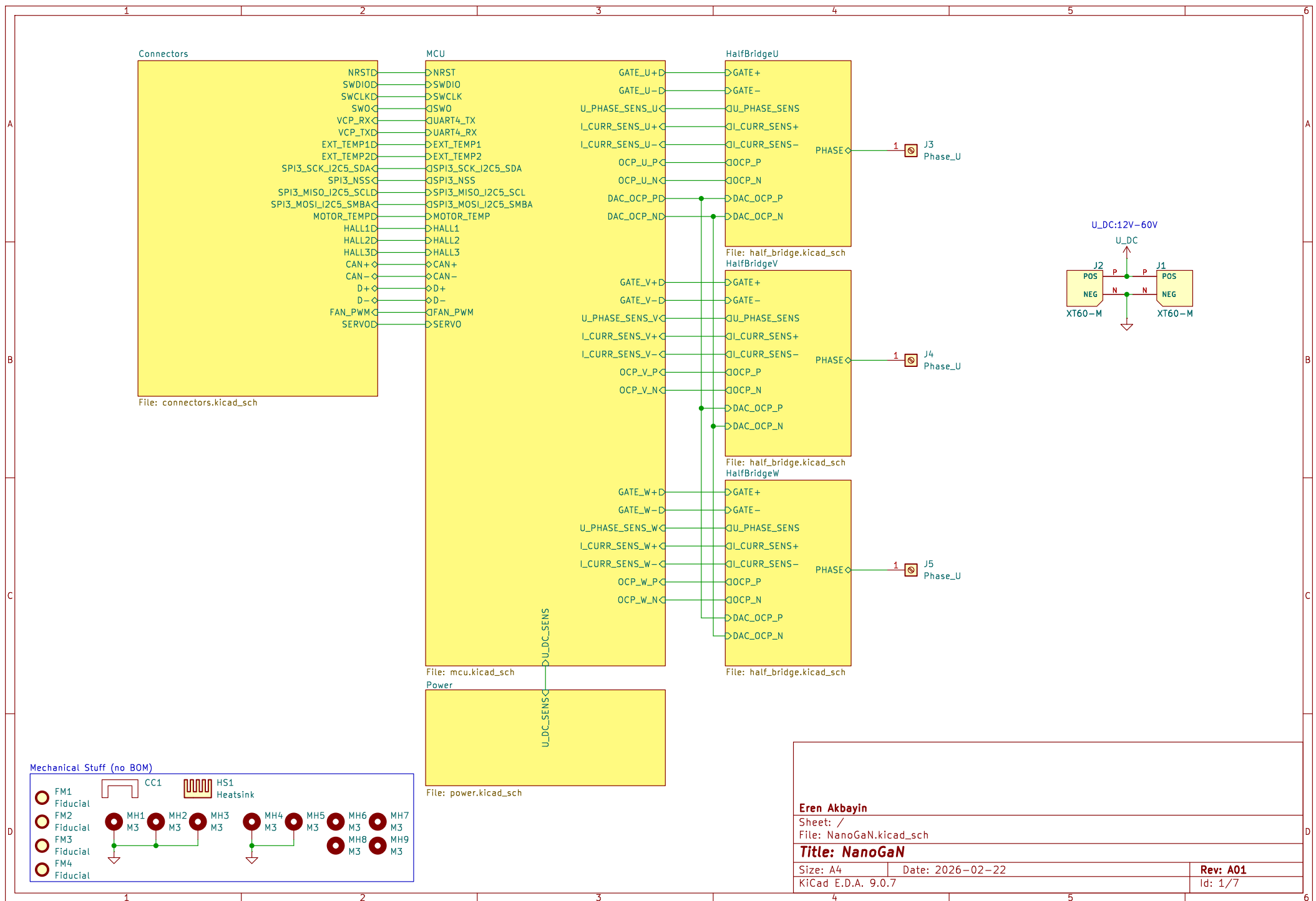
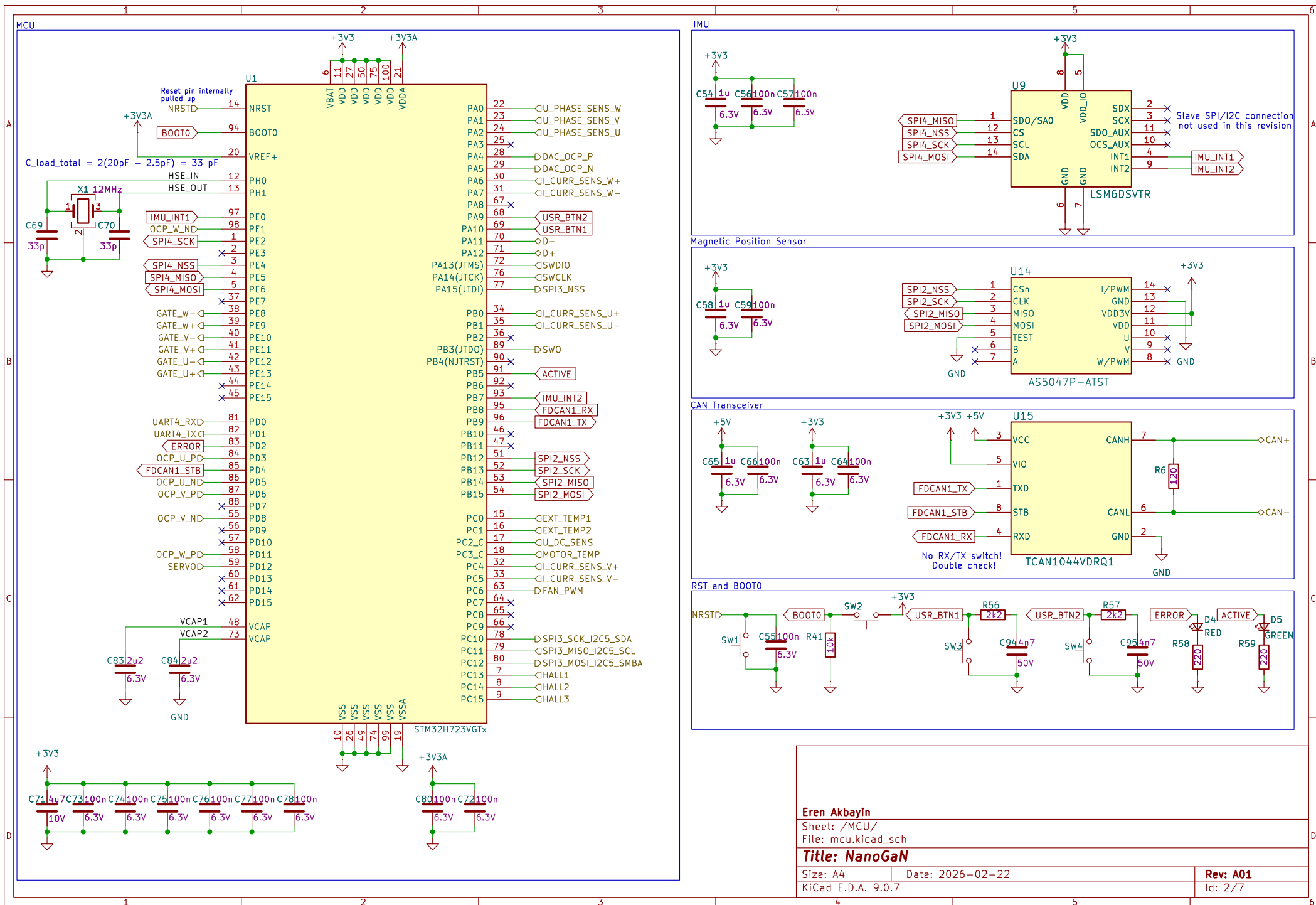






Eren Akbayin

Sheet: /MCU/
File: mcu.kicad_sch

Title: NanoGaN

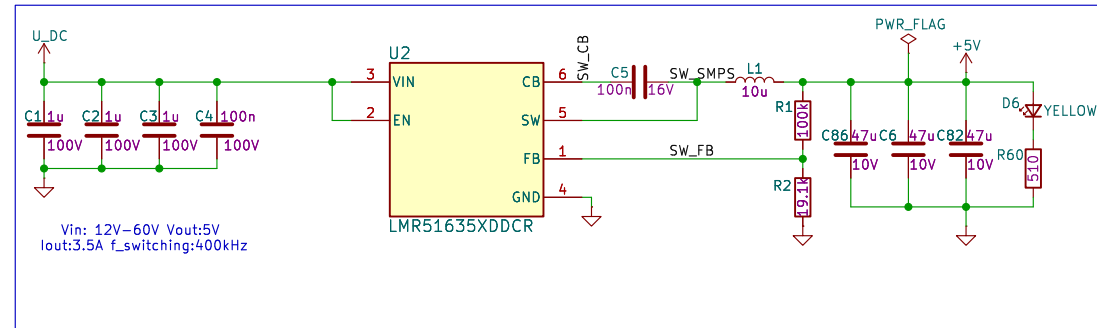
Size: A4
KiCad E.D.A. 9.0.7

Date: 2026-02-22

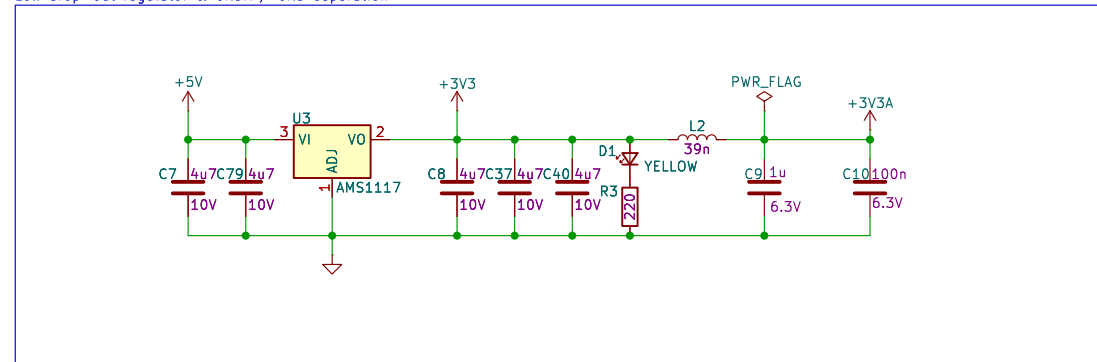
Rev: A01

Id: 2/7

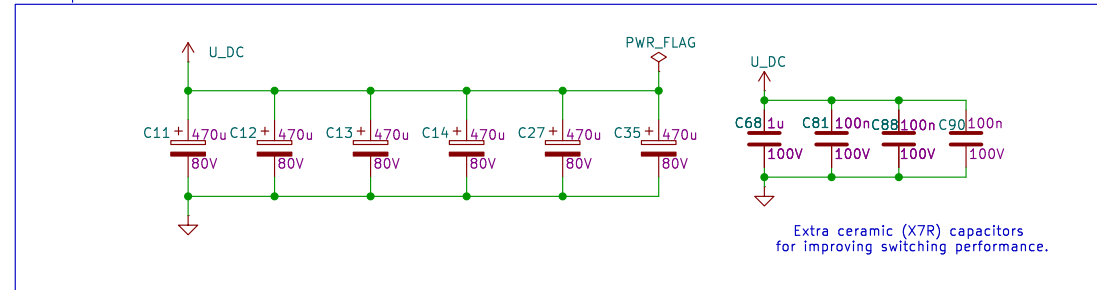
Buck Converter



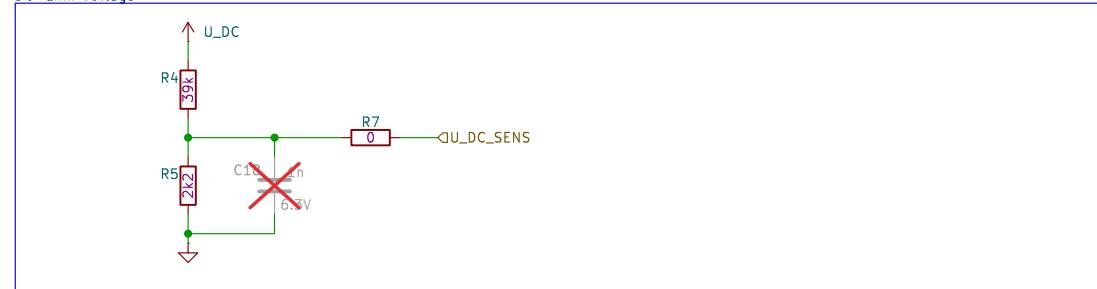
Low drop-out regulator & GNDA / GND separation



Bulk Capacitors



DC-Link Voltage

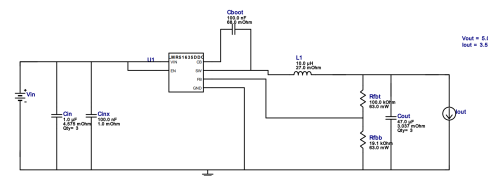


WEBENCH[®] Design Report

Design : 29 LMR51635DDC
LMR51635DDC 12V-60V to 5.00V @ 3.5A

VinMin = 12.0V
VinMax = 60.0V
Vout = 5.0V
Iout = 3.5A

Device = LMR51635DDC
Topology = Buck
Created = 2026-01-18 17:57:50.631
BOM Cost = \$2.43
BOM Count = 12
Total Pd = 2.47W



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Sheet: /Power/
File: power.kicad_sch

Title: NanoGaN

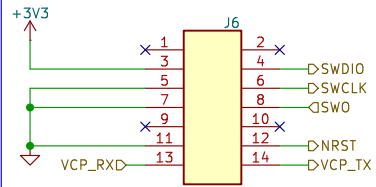
Size: A4	Date: 2026-02-22
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KiCad E.D.A. 9.0.7

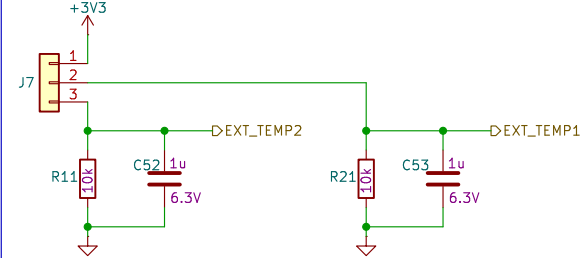
Rev: A01

Id: 3/7

ST-Link V3 MINIE CONNECTOR (STDC14)

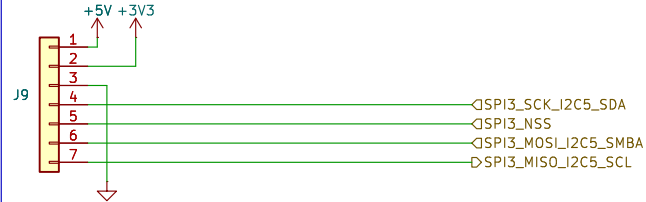


External Temp. Readings

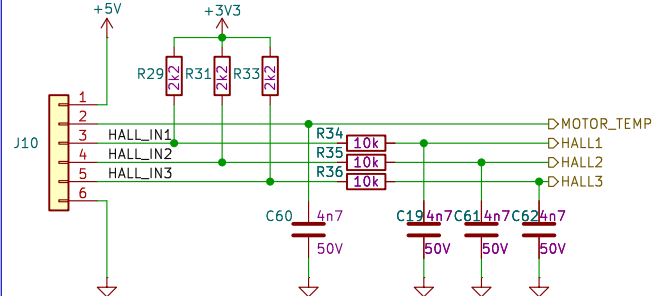


for motor and etc.

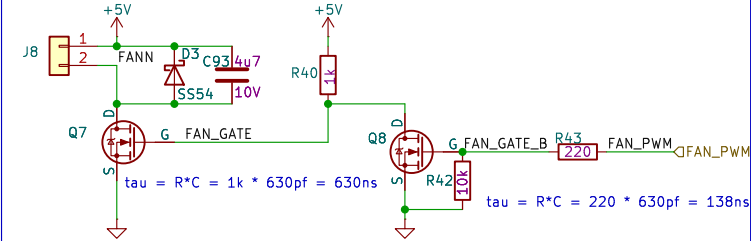
SPI/I2C/USART connection



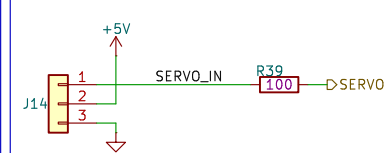
Hall connector and filtering



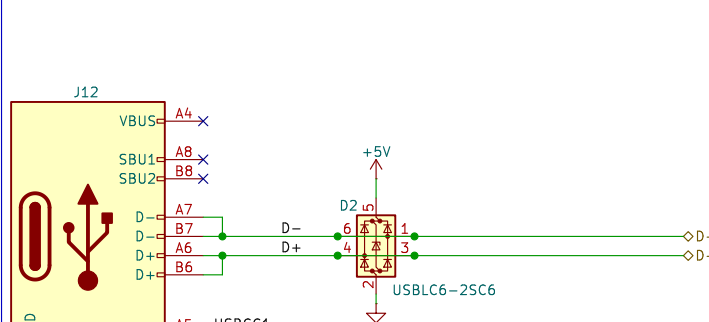
Fan connector



Servo



USB-C Connector



Differential Pair settings

Layers	4	PCB Thickness	0.8	Inner Copper Weight	0.3oz	Outer Copper Weight	1oz	Unit	mm
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Impedance Configure								
Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Spacing (mm)	Trace Width (mm)	Impedance trace to copper (mm)	
90	Differential Pair (Non-coupled)	L1	-	L2	-	0.20	0.20	-
100	Differential Pair (Non-coupled)	L8	-	L2	-	0.5	0.5	-

JLC0808H-2310 (Standard) (Finished thickness: 0.74±0.10mm)								
Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width (mm)	Trace Spacing (mm)	Impedance trace to copper (mm)	
L1	Outer Copper Weight		1.35		0.20			
Prepreg	0.075mm VMCZ without copper		3.91		0.0994			
L2	Inner Copper Weight		0.60		0.0750			
Core	0.075mm VMCZ without copper		2.95		0.0750			
L3	Inner Copper Weight		0.60		0.0994			
Prepreg	0.075mm VMCZ without copper		3.91		0.0994			
L4	Inner Copper Weight		0.60		0.0750			
Core	0.075mm VMCZ without copper		2.95		0.0750			
L5	Inner Copper Weight		0.60		0.0994			
Prepreg	0.075mm VMCZ without copper		3.91		0.0994			
L6	Outer Copper Weight		1.35		0.20			

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Sheet: /Connectors/
File: connectors.kicad_sch

Title: NanoGaN

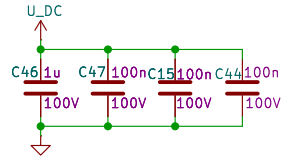
Size: A4 Date: 2026-02-22

KiCad E.D.A. 9.0.7

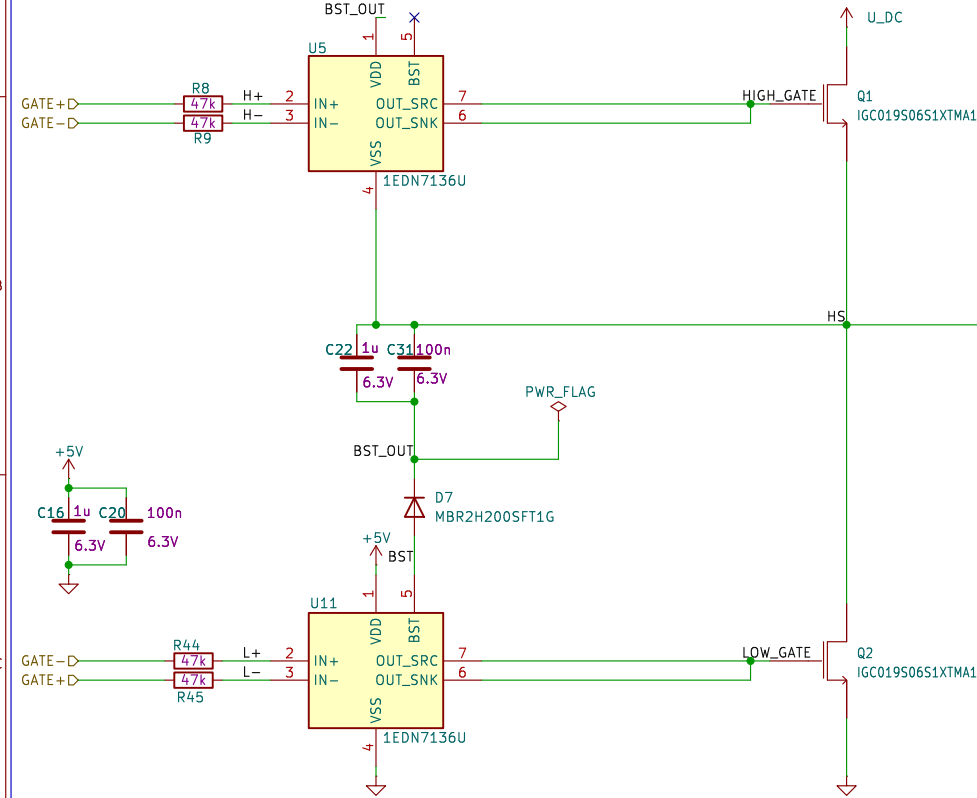
Rev: A01

Id: 4/7

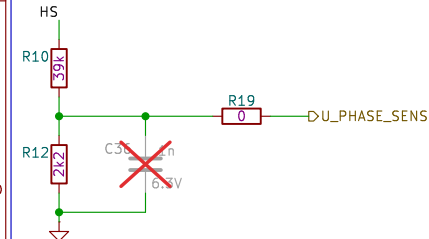
U_DC High Frequency Decoupling Capacitors



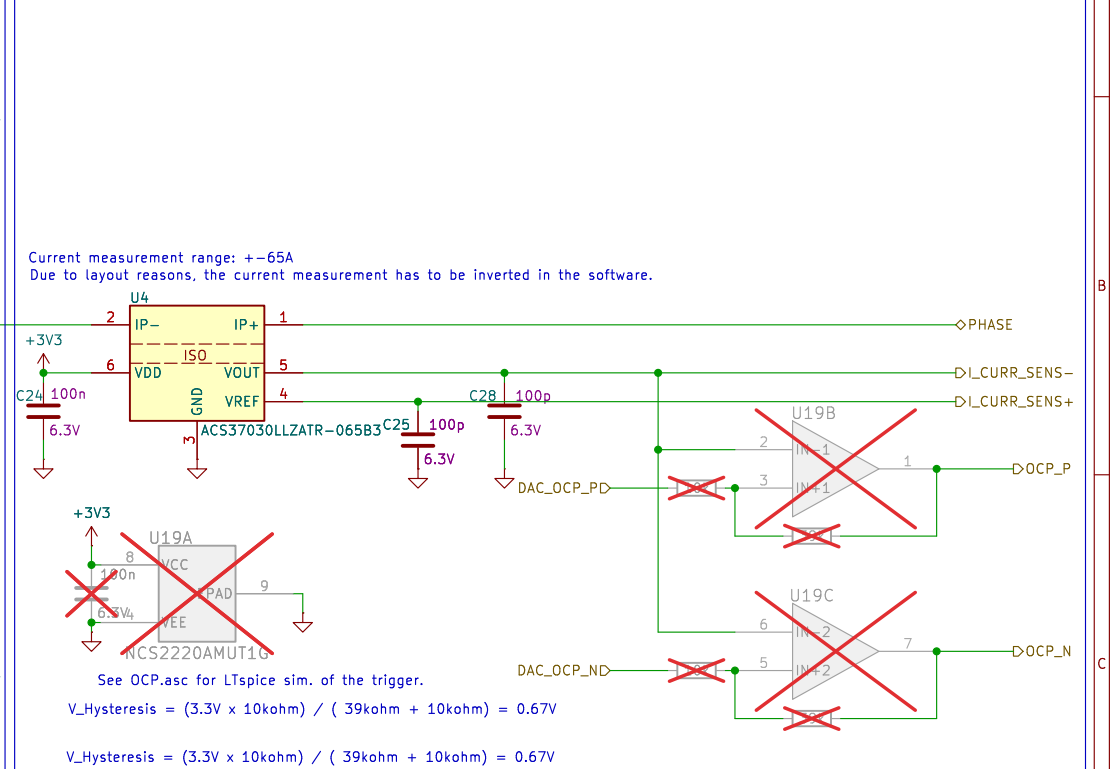
Halfbridge and GateDrivers



Voltage Measurement



Current Measurement



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Sheet: /HalfBridgeU/
File: half_bridge.kicad_sch

Title: NanoGaN

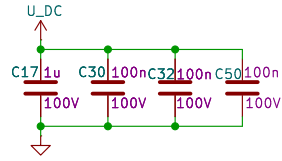
Size: A4 Date: 2026-02-22

KiCad E.D.A. 9.0.7

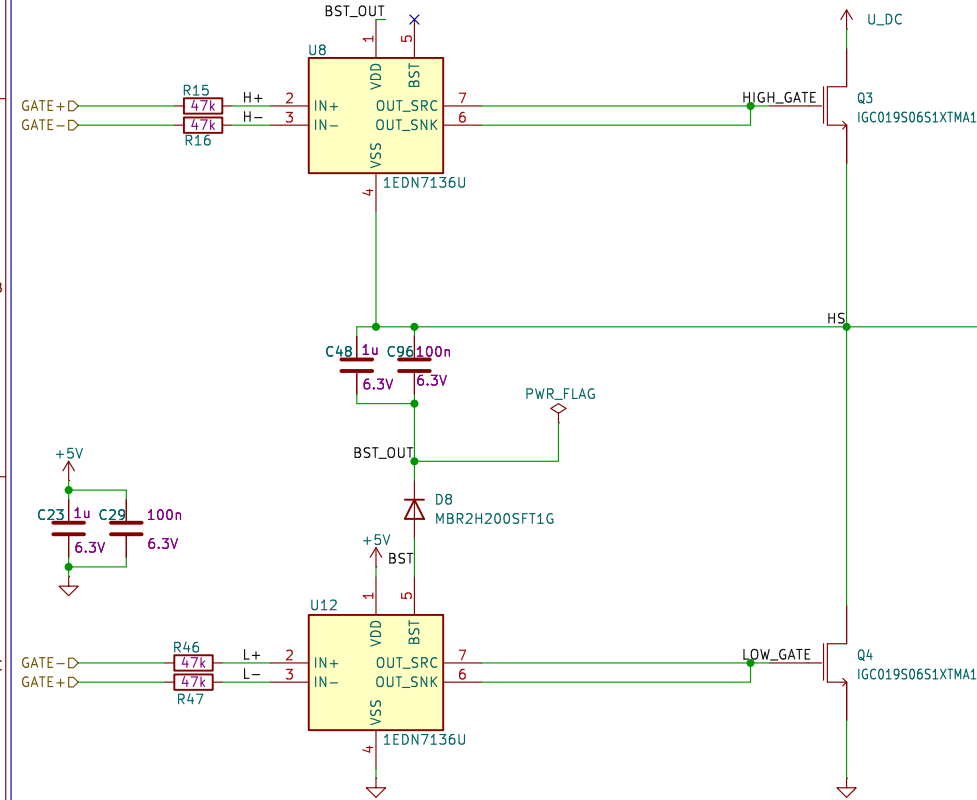
Rev: A01

Id: 5/7

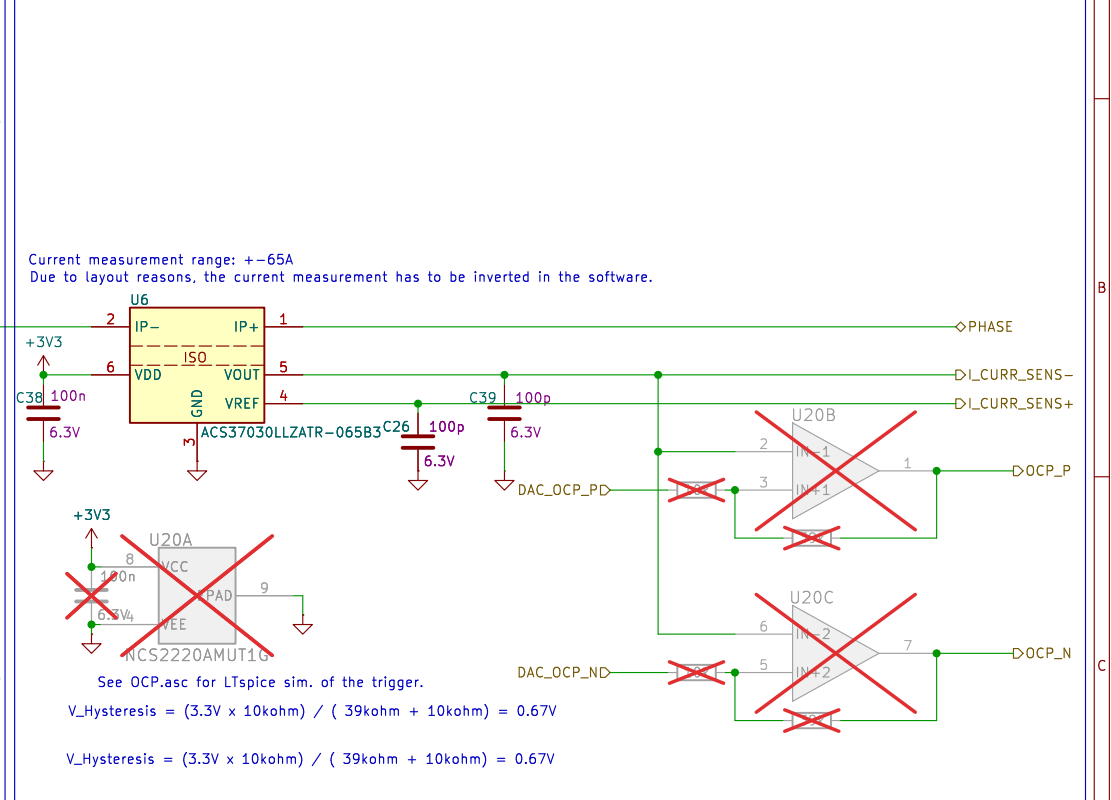
U_DC High Frequency Decoupling Capacitors



Halfbridge and GateDrivers



Current Measurement



Voltage Measurement



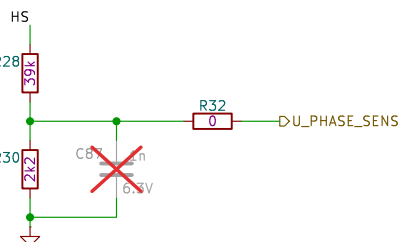
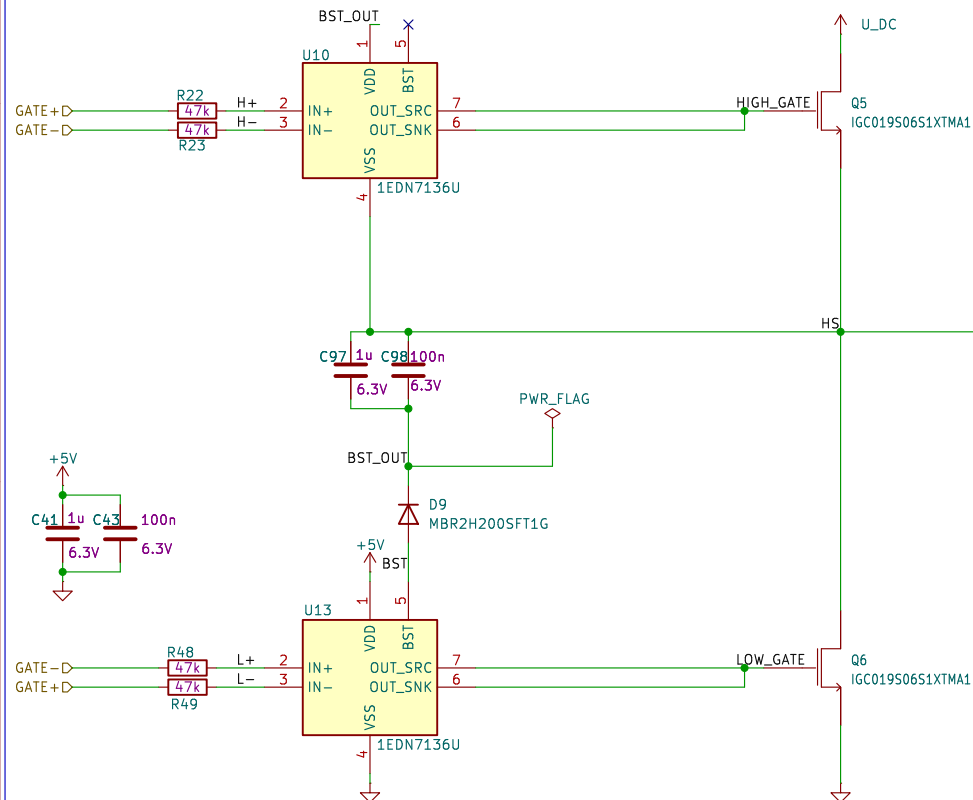
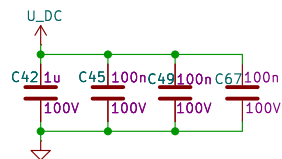
Eren Akbayin

Sheet: /HalfBridgeV/
File: half_bridge.kicad_sch

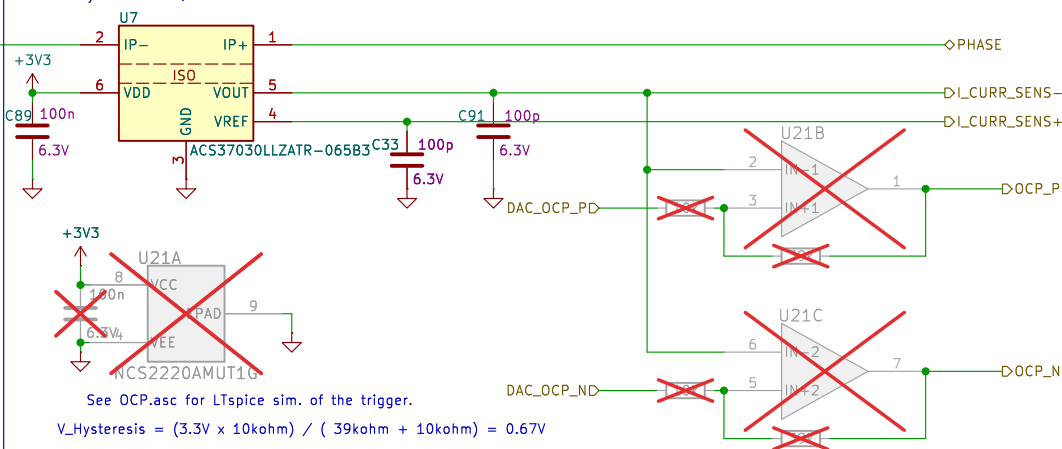
Title: NanoGaN

Size: A4 Date: 2026-02-22
KiCad E.D.A. 9.0.7

Rev: A01
Id: 6/7



Current measurement range: $\pm 65\text{A}$
Due to layout reasons, the current measurement has to be inverted in the software.



See OCP.asc for LTspice sim. of the trigger.

$$V_{\text{Hysteresis}} = (3.3\text{V} \times 10\text{kohm}) / (39\text{kohm} + 10\text{kohm}) = 0.67\text{V}$$

$$V_{\text{Hysteresis}} = (3.3\text{V} \times 10\text{kohm}) / (39\text{kohm} + 10\text{kohm}) = 0.67\text{V}$$

Sheet: /HalfBridgeW/
File: half_bridge.kicad_sch

Size: A4	Date: 2026-02-22
KiCad E.D.A. 9.0.7	

Rev: A01
Id: 7/7